

Intake Manifold 1 Signal Malfunction

Symptoms

- The OEM charge air temperature (left bank rear) sensor has malfunctioned.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The charge air temperature (left bank rear) sensor is connected to the Alarm and Safety C4 connector located on the customer interface box.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box wiring.	
	STEP 1A. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for an open.	
	STEP 1B. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for a wire-to-wire short.	
	STEP 1C. Check the charge air temperature (left bank rear) signal wire for a short to ground.	
STEP 2.	Check the OEM wiring harness.	

STEPS		SPECIFICATIONS
	STEP 2A. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for an open.	
	STEP 2B. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for a wire-to-wire short.	
	STEP 2C. Check the charge air temperature (left bank rear) signal wire for a short to ground.	

STEP 1. Check the customer interface box wiring.

STEP 1A. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for an open.

Conditions : • Open the customer interface box. • Disconnect the charge air temperature (left bank rear) signal, return, and return 2 wires at the remote input/output unit. • Disconnect the C4 connector.		
Action	Specification/Repair	Next Step

Check the charge air temperature (left bank rear) signal, return, and return 2 wires for an open. NOTE: An alarm will sound on the remote input/output unit when an open is detected.	Less than 10 ohms? YES	1B
• Place one test lead on the charge air temperature (left bank rear) signal wire at the remote input/output unit. • Place the other test lead on the charge air temperature (left bank rear) signal pin at the C4 connector. • Place one test lead on the charge air temperature (left bank rear) return wire at the remote input/output unit. • Place the other test lead on the charge air temperature (left bank rear) return pin at the C4 connector. • Place one test lead on the charge air temperature (left bank rear) return 2 wire at the remote input/output unit. • Place the other test lead on the charge air temperature (left bank rear) return 2 pin at the C4 connector.		

Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.

Less than 10 ohms?

NO

Repair :

Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15.
(/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)

Repair complete

STEP 1B. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the charge air temperature (left bank rear) signal, return, and return 2 wires at the remote input/output unit.
- Disconnect the C4 connector.

Action	Specification/Repair	Next Step

Check the charge air temperature (left bank rear) signal, return, and return 2 wires for a wire-to-wire short. • Place one test lead on the charge air temperature (left bank rear) signal wire at the remote input/output unit. • Place the other test lead on all other wires at the remote input/output unit. • Place one test lead on the charge air temperature (left bank rear) return wire at the remote input/output unit. • Place the other test lead on all other wires at the remote input/output unit. • Place one test lead on the charge air temperature (left bank rear) return 2 wire at the remote input/output unit. • Place the other test lead on all other wires at the remote input/output unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	1C

STEP 1C. Check the charge air temperature (left bank rear) signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the charge air temperature (left bank rear) signal wire at the remote input/output unit.
- Disconnect the C4 connector.

Action	Specification/Repair	Next Step
Check the charge air temperature (left bank rear) signal wire for a short to ground. • Place one test lead on the charge air temperature (left bank rear) signal wire at the remote input/output unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check the OEM wiring harness.**STEP 2A. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for an open.****Conditions :**

- Disconnect the OEM harness at the C4 and C11 connectors.
- Disconnect the charge air temperature (left bank rear) sensor connector.

Action	Specification/Repair	Next Step

Check the charge air temperature (left bank rear) signal, return, and return 2 wires for an open.

NOTE: An open has been detected by the remote input/output unit if a false alarm has occurred.

- Place one test lead on the charge air temperature (left bank rear) signal pin at the C4 connector.
- Place the other test lead on the charge air temperature (left bank rear) signal pin at the C11 connector.
- Place one test lead on the charge air temperature (left bank rear) return pin at the C4 connector.
- Place the other test lead on the charge air temperature (left bank rear) return pin at the C11 connector.
- Place one test lead on the charge air temperature (left bank rear) return 2 pin at the C4 connector.
- Place the other test lead on the charge air temperature (left bank rear) return 2 pin at the C11 connector.
- Place one test lead on the charge air temperature (left bank

Less than 10 ohms?

YES

2B

rear) signal pin at the C11 connector.

- Place the other test lead on the charge air temperature (left bank rear) signal pin at the sensor connector.
- Place one test lead on the charge air temperature (left bank rear) return pin at the C11 connector.
- Place the other test lead on the charge air temperature (left bank rear) return pin at the sensor connector.
- Place one test lead on the charge air temperature (left bank rear) return 2 pin at the C11 connector.
- Place the other test lead on the charge air temperature (left bank rear) return 2 pin at the sensor connector.

Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.

Less than 10 ohms?

NO

Repair :

Replace the wire. Refer to the OEM service manual.

Repair complete

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STEP 2B. Check the charge air temperature (left bank rear) signal, return, and return 2 wires for a wire-to-wire short.

Conditions : • Disconnect the OEM harness at the C4 and C11 connectors. • Disconnect the charge air temperature (left bank rear) sensor connector.		
Action	Specification/Repair	Next Step

Check the charge air temperature (left bank rear) signal, return, and return 2 wires for a wire-to-wire short. • Place one test lead on the charge air temperature (left bank rear) signal pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the charge air temperature (left bank rear) return pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the charge air temperature (left bank rear) return 2 pin at the C4 connector. • Place the other test lead on all other pins at the C4 connector. • Place one test lead on the charge air temperature (left bank rear) signal pin at the C11 connector. • Place the other test lead on all other pins at the C11 connector. • Place one test lead on the charge air temperature (left bank rear) return pin at the C11 connector.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
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- Place the other test lead on all other pins at the C11 connector. - Place one test lead on the charge air temperature (left bank rear) return 2 pin at the C11 connector. - Place the other test lead on all other pins at the C11 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	2C
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STEP 2C. Check the charge air temperature (left bank rear) signal wire for a short to ground.

Conditions : - Disconnect the OEM harness at the C4 and C11 connectors. - Disconnect the charge air temperature (left bank rear) sensor connector.		
Action	Specification/Repair	Next Step

Check the charge air temperature (left bank rear) signal wire for a short to ground. - Place one test lead on the charge air temperature (left bank rear) signal pin at the C4 connector. - Place the other test lead on engine ground. - Place one test lead on the charge air temperature (left bank rear) signal pin at the C11 connector. - Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO Repair : Replace the charge air temperature (left bank rear) sensor. Contact a Cummins® Authorized Repair Location.	Repair complete